

WEIYANG HUANG

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University of Washington, Seattle, WA 98105, USA

Research Interests: Multimodal learning, 3D visual reasoning, visual grounding, compositional generalization, and reliable evaluation of vision-language models.

Education

University of Washington <i>Master of Science in Electrical and Computer Engineering (PMP)</i>	Sep. 2026 - Present
Wuhan Textile University <i>Bachelor of Engineering in Electrical Engineering and Automation</i> <i>Bachelor of Economics in Finance (Dual Degree)</i> GPA: 3.78/4.00	Sep. 2022 - Jun. 2026

Professional Experience

Research Assistant Study on Continuous Learning and Catastrophic Interference - Developed a deep learning framework for continual learning, introducing a multi-prototype mechanism to improve model stability and generalization. - Designed feature expansion and data-generation components, and reconstructed the training pipeline to support multi-task learning under contrastive loss. - Investigated continual learning and catastrophic interference through connections to cognitive and human memory studies.	Apr. 2025 - Dec. 2025
Research Assistant Electrical Laboratory, Wuhan Textile University - Studied CNN architectures and applications for image analysis and NLP, including GhostNet and ResNet18, with emphasis on efficient inference. - Introduced MobileNeXT-inspired lightweight modeling and evaluated networks using IoU, F1, and related performance metrics. - Designed and tested applications on resource-constrained devices to validate the usability of lightweight CNNs.	Feb. 2025 - Nov. 2025
Intern Extra-High Voltage Branch, State Grid Corporation of China - Contributed to software and hardware design for automated drone flights in powerline inspection, improving autonomous fault detection in complex environments. - Integrated image processing and adaptive path planning to improve inspection efficiency and reliability; supported lightning-protection risk analysis using route geography. - Coordinated logistics, documentation, and reception for a national insulator academic seminar.	Jul. 2025 - Aug. 2025
Researcher Power System Load Flow Analysis - Simulated power-system operation and solved load-flow equations with the Newton-Raphson method, assessing feasibility and convergence. - Built a digital model in MWorks with synchronous generators, transformers, and parallel RLC components; implemented sensitivity and reliability analyses.	Sep. 2024 - Dec. 2024
Researcher Product Optimization with MDP and MCTS - Developed Markov decision process models for manufacturing optimization, balancing quality control, cost minimization, and uncertainty. - Derived Monte Carlo tree search algorithms and evaluated defect-rate and inspection-cost sensitivities to improve robust design.	Sep. 2024
Intern Yichang Yineng Hydropower Co., Ltd. - Monitored hydropower-generator operation, including wicket-gate position and penstock pressure, and supported routine turbine and control-system maintenance. - Analyzed SCADA data to identify software and hardware issues and develop performance-optimization and preventive-maintenance strategies.	Jan. 2024 - Feb. 2024
Researcher Deep Learning-based Facial Recognition System - Contributed to a facial-recognition system for ATM authentication, building robust feature-extraction components for variable lighting and multiple datasets. - Implemented core features, tested model components, and supported debugging and performance refinement.	Feb. 2023 - Jun. 2023

Selected Self-Study Coursework

- **Stanford CS231n: Convolutional Neural Networks for Visual Recognition** - CNN architectures, transfer learning, object detection, semantic segmentation, and visual recognition.
- **Stanford CS224N: NLP with Deep Learning** - word vectors, sequence-to-sequence models, attention, Transformers, and foundations of large language models.
- **Stanford CS230: Deep Learning** - optimization, regularization, model training, error analysis, and structuring machine-learning projects.
- **Stanford CS234: Reinforcement Learning** - Markov decision processes, value and policy methods, exploration, and deep reinforcement learning.

Activities and Leadership

Member	Stray Cat Protection Association	Sep. 2024 - Sep. 2025
Director	Electronic Design Association, Wuhan Textile University	Sep. 2023 - Jun. 2024
Volunteer	Volunteer Association, Wuhan Textile University	Jul. 2023 - Aug. 2023

Skills

Programming: Python, C++, Java, MATLAB, Stata, SQL
Machine Learning & Computer Vision: PyTorch, CNNs, Continual Learning, Image Processing, Reinforcement Learning
Tools & Systems: Git, Linux, MWorks, SCADA
Languages: Mandarin (Native), English (IELTS 7.5; GRE 325 + 3.5)

Honors and Awards

Outstanding Graduate (Top 1%)	Jun. 2026
First-class Scholarship Wuhan Textile University	Sep. 2023; Oct. 2024; Oct. 2025
Second Prize Contemporary Undergraduate Mathematical Contest in Modeling	Sep. 2024
Second Prize Siemens Cup Smart Manufacturing Challenge, China	Jul. 2024
Excellent Award; Second Prize Blue Bridge Cup National Software and IT Professional Talent Competition	May 2023; Dec. 2024